THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is the effect of atmospheric refraction? 1

(a) Rainbow **(b)** Advance sunrise **(c)** Tyndall effect **(d)** Blue sky

2. A normal eye can see 1

(a) only objects at infinity **(b)** from 1 m to 2 m

(c) from 0 to 25 cm **(d)** from 25 cm to infinity

3. A narrow beam of white light is passed through a glass prism. A spectrum is NOT formed if the beam is replaced by 1

(a) a beam of pure red light **(b)** sunlight

(c) light from a tube **(d)** light from a bulb

4. Smoke is used in a room to see a beam of light because 1

(a) smoke produces light **(b)** smoke reflects colours

(c) smoke particles scatter light **(d)** smoke is bright

5. Which defect of vision usually develops with age, even in a person of normal vision earlier? 1

(a) Colour blindness **(b)** Cataract only **(c)** Myopia only **(d)** Presbyopia

6. Light of which colour has the longest wavelength and is scattered the least? 1

(a) Blue **(b)** Violet **(c)** Red **(d)** Green

7. Fog lamps have yellow or red light because 1

(a) it is scattered less **(b)** it is cooler

(c) it is cheaper **(d)** it is brighter

8. A person with a near point of 100 cm needs spectacles of 1

(a) convex lenses of power +3 D **(b)** concave lenses of -1 D

(c) concave lenses of -3 D **(d)** convex lenses of +1 D

9. When light passes through a rectangular slab, it shows 1

(a) no dispersion **(b)** absorption

(c) dispersion in seven colours **(d)** total internal reflection

10. Which two colours are most scattered in the atmosphere? 1

(a) Red and orange **(b)** Green and yellow **(c)** Violet and blue **(d)** Red and violet

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a person with myopia advised not to use a convex lens? 2

12. Why does the focal length of the eye lens decrease when we look at a nearby object? 2

13. Why does the Sun appear reddish when it is near the horizon? 2

14. Why does the Sun look flattened at sunrise? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How would you correct a person with both myopia and hypermetropia? 3

16. Describe the role of the iris and the pupil in protecting the retina. 3

17. Why do some persons see blurred images in old age? How can it be corrected? 3

OR

Distinguish between dispersion and scattering with one example each. 3

18. What would happen to vision if the eyeball were slightly longer than the usual size? Name the defect and the correction. 3

OR

Why is blue light not used for danger signals even if it is bright? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rainbow is observed in the morning, with the Sun in the east. In which direction is the observer facing and why? Explain the formation of a rainbow. 5

OR

A student can read a book at 25 cm but cannot see the board 6 m away clearly. Diagnose the defect, state the cause, find the focal length and power of the lens (far point 6 m). 5

20. Explain with reasons: 5

(a) the sky is blue

(b) the clouds are white

(c) the Sun looks red at the horizon.

OR

Distinguish between the Tyndall effect and dispersion. 5

END OF QUESTION PAPER — Total: 40 Marks